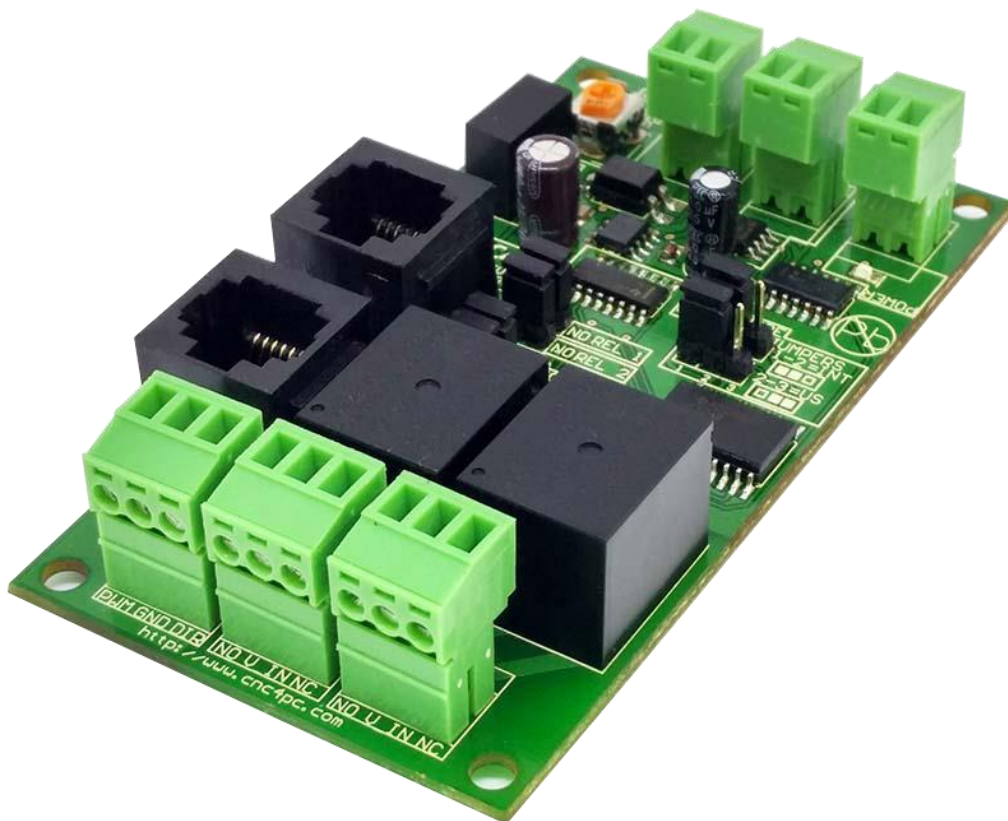


**C41S – VARIABLE SPEED
CONTROL Rev. 1.1**



APRIL 2018

USER'S MANUAL

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1.0 OVERVIEW

This card lets you control your spindle with PWM and direction signals, as if it was an axis motor. It converts the step signal into an analog (0-10VDC). This card also has two relays that can be used to control the direction (CW/CCW) and enable the drive (On/Off). A Variable Frequency Drive or Inverter works by modifying the frequency for AC motors. Most of these devices can be controlled with an external analog signal (0-10VDC). That is, if there is 5VDC coming in through the control signal, the motor will run at 50% of full speed, if there was 10VDC, the motor will run at 100% of full speed. If there is no signal coming out, then the motor will stop.

This unit can also be used on many DC motor controllers by replacing the potentiometer that controls the speed.

This version can be easily mounted on control boxes using DIN rails and can also accommodate ribbon cables or DB25 connectors.

2.0 FEATURES

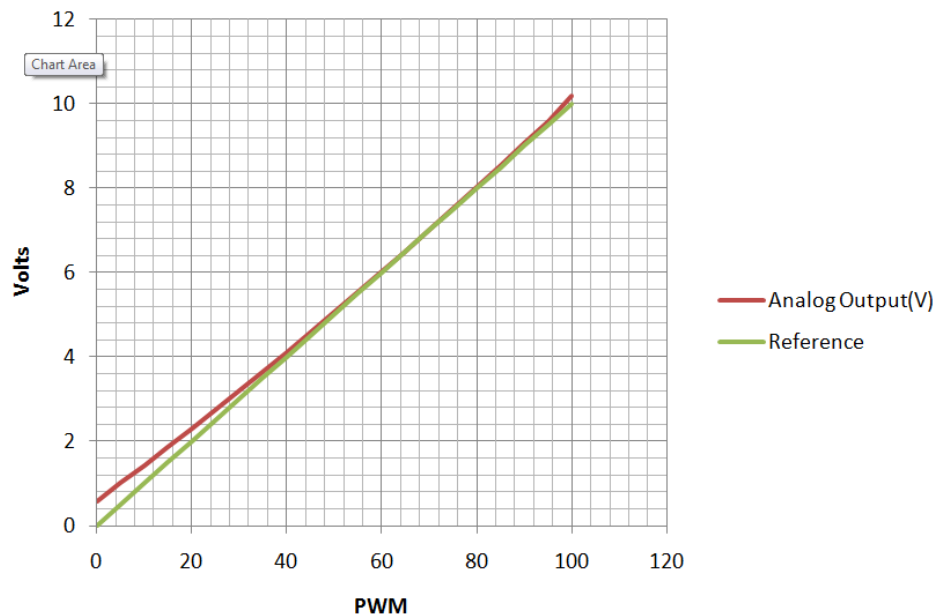
- **Inputs a PWM signal and outputs an analog 0-10VDC.**
- **Uses only two pins, one for PWM and one for direction.** It is the presence or absence of the valid PWM signal that would start/stop the spindle.
- **Has two relays that can be used to control the direction and enable and disable the drive.**
- **Jumper to select mode of operation, US or International mode.** On US mode, one relay would be used to start CW and the second one to start CCW. On International mode Relay 1 will start/stop, and Relay 2 will determine the direction of rotation.
- **Optoisolated output signals.** The analog and CW/CCW signals are optically isolated, so this board can be used with drives that make grounds common with the mains that drive the VFD or motor.
- **All TTL +5VDC Signals.** Interface directly with parallel port interface products and other cnc4pc.com cards. 5VDC (TTL) signals are very common among automation devices.
- **Screw-On connections for all terminals.** You only have to screw-on the wires to make all your connections.
- **Flexible design.** It works with cnc4pc's products, directly through your parallel port, or through many other parallel port control products.
- **Din rail mountable. *NEW*.**
- **Pluggable Screw-On Terminals. *NEW*.**

3.0 SPECIFICATIONS

INPUT SPECIFICATIONS	
On-state voltage range	2 to 5V DC
Minimum on-state input current	10 mA
Recommended PWM signal frequency	(200Hz)
Signal Type	Active High

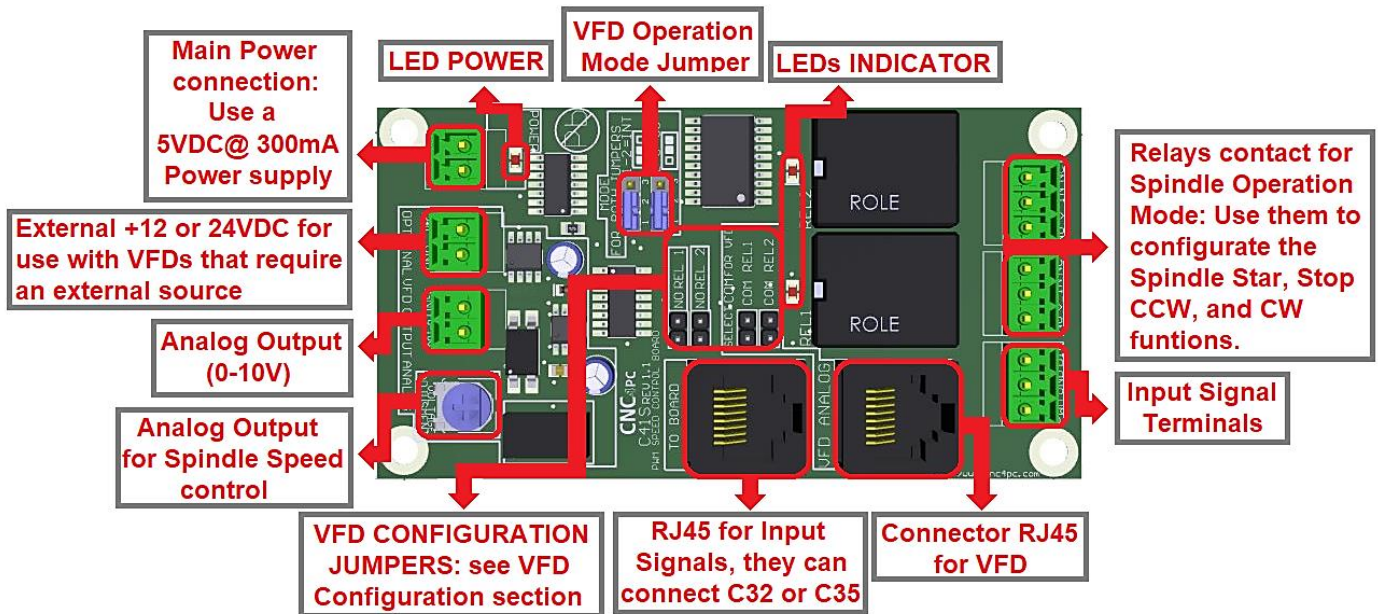
ELECTROMECHANICAL RELAYS SPECIFICATIONS	
Maximum Current (AC)	7A@240VAC; 10A@125VAC
Maximum Current (DC)	15A@24VDC; 10A@28VDC

GENERAL SPECIFICATIONS	
Dimensions	Millimeters: 105L x 75W x22D
External power	5V DC @ 300mA



Linearity Analysis (Analog voltage vs. %PWM)

4.0 BOARD DESCRIPTION



Requirements:

It requires a +5VDC@300mA power supply to operate.

Note: this connection is common to the PC GND

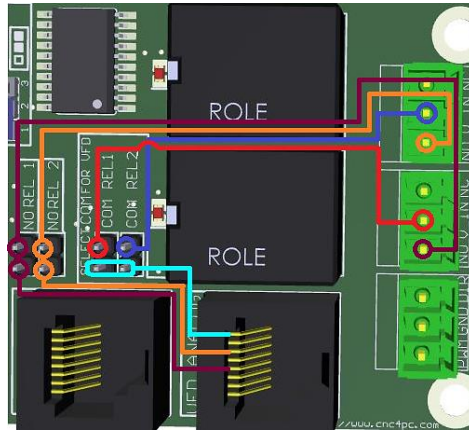


WARNING: To keep the output signals optoisolated, these must not have common ground or current with other circuits you are using.

You will require a voltmeter to fine tune your system.

5.0 RELAY AND JUMPER INTERNAL WIRING.

In this diagram shows connections internal of the relays with the jumper and the RJ45 connector for the VFD



6.0 RJ45 FOR VFD CONNECTION

The RJ45 allows connecting to a VFD using a CAT5 cable from the C41S to pigtails on the other end of the wire to the terminals of the VFD.

RJ45 for VFD	
RJ45 PIN	Function
1	An. GND
2	Analog Output
3	Not Used
4	REL 1 N.O. Contact
5	Ext. GND
6	REL 2 N.O. Contact
7	Ext. 12VDC or 24VDC
8	Relay Common

An. GND: Ground of the Analog output signal

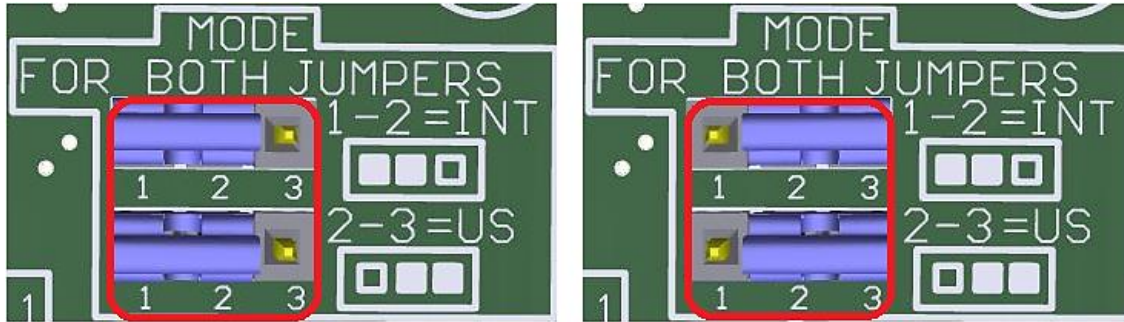
Analog Output: Isolated Analog Output Signal (0-10V)

Ext. GND: External 12V or 24V power supply GND.

Ext. 12VDC or 24VDC: External 12VDC or 24VDC power supply for using it with VFDs that require an external power source.

Relay Common: The signal or voltage wired to this terminal can be connected to the common terminals of the relay 1 and relay 2. Use the on-board RELAY COMMON JUMPERS to do this connection. Remove the jumper if this connection is not required.

7.0 OPERATION MODE JUMPER



Operation mode jumpers

This jumper allows selecting the way how the relays go to be activated when a PWM signal and REV signal are present in the inputs terminals. See the tables below.

AMERICAN MODE (US)			
INPUTS		RELAYS	
<i>PWM</i>	<i>REV</i>	<i>REL 1</i>	<i>REL 2</i>
ON	ON	OFF	ON
ON	OFF	ON	OFF
OFF	ON	OFF	OFF
OFF	OFF	OFF	OFF

INTERNATIONAL MODE (INT)			
INPUTS		RELAYS	
<i>PWM</i>	<i>REV</i>	<i>REL 1</i>	<i>REL 2</i>
ON	ON	ON	ON
ON	OFF	ON	OFF
OFF	ON	OFF	OFF
OFF	OFF	OFF	OFF

Relay 1 and 2

They can be used to control the VFD. The specification of the relays are:

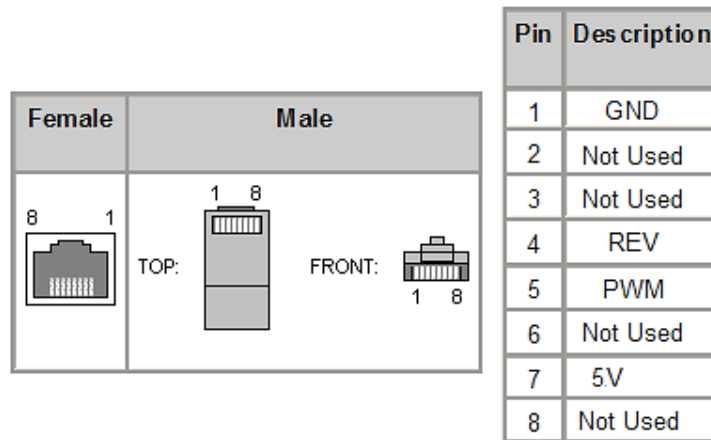
ELECTROMECHANICAL RELAYS SPECIFICATIONS	
Maximum Current (AC)	7A@240VAC; 10A@125VAC
Maximum Current (DC)	15A@524VDC; 10A@28VDC

Electromechanical Relays Specifications

8.0 RJ45 FOR INPUT SIGNAL

The RJ45 connector is for connecting the boards directly using a standard Ethernet cable. A standard Ethernet cable with RJ45 has straight through wiring. That means that Pin1 = Pin1, etc....

This connector eases the installation process by eliminating the screw on terminals. The RJ45 connector is used not just to carry the I/O signals placed in terminals, but also to power board.



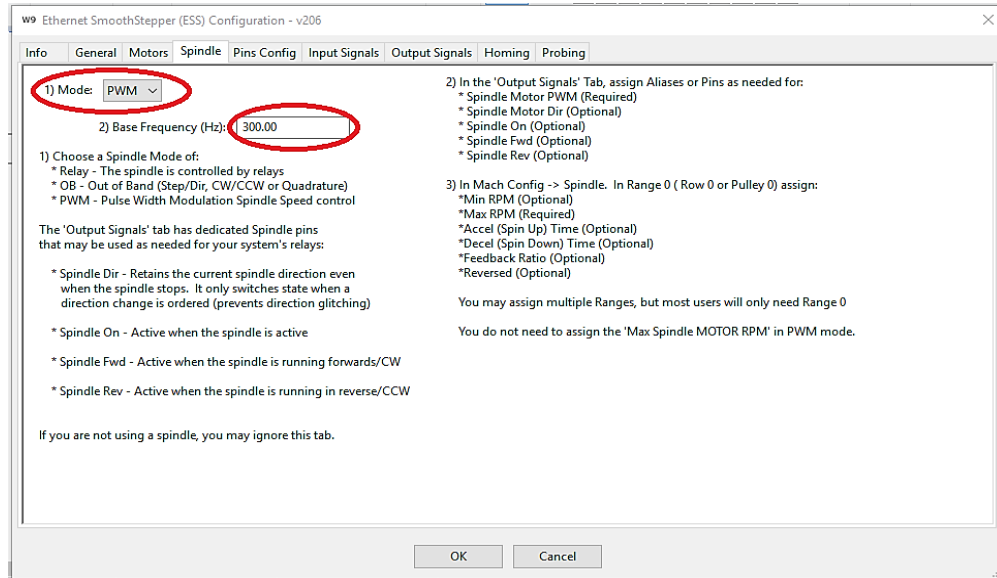
Before connecting anything, please be sure to read your VFD's manual and make sure you understand all the safety issues.

9.0 CONFIGURING THE CONTROL SOFTWARE:

It is strongly recommended you read the control software's manual. You need to configure your control software to control the spindle as if it was an angular axis. This card requires a PWM input signal to deliver 10VDC. So, you have to set the speed of the motor (spindle) at maximum. For acceleration values adjust them to where you feel comfortable. Keep in mind the acceleration of the motor must also be set in your VFD.

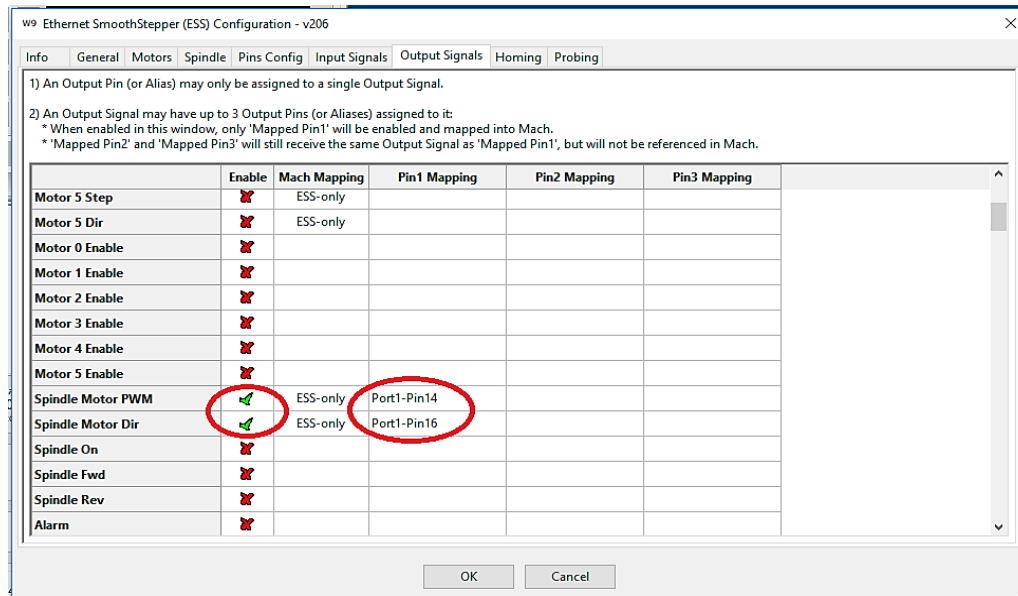
9.1 For configuring Mach4 follow these steps:

1. Go to Configure / Plugins / ESS / Spindle. Active the mode in PWM and select the Frequency.



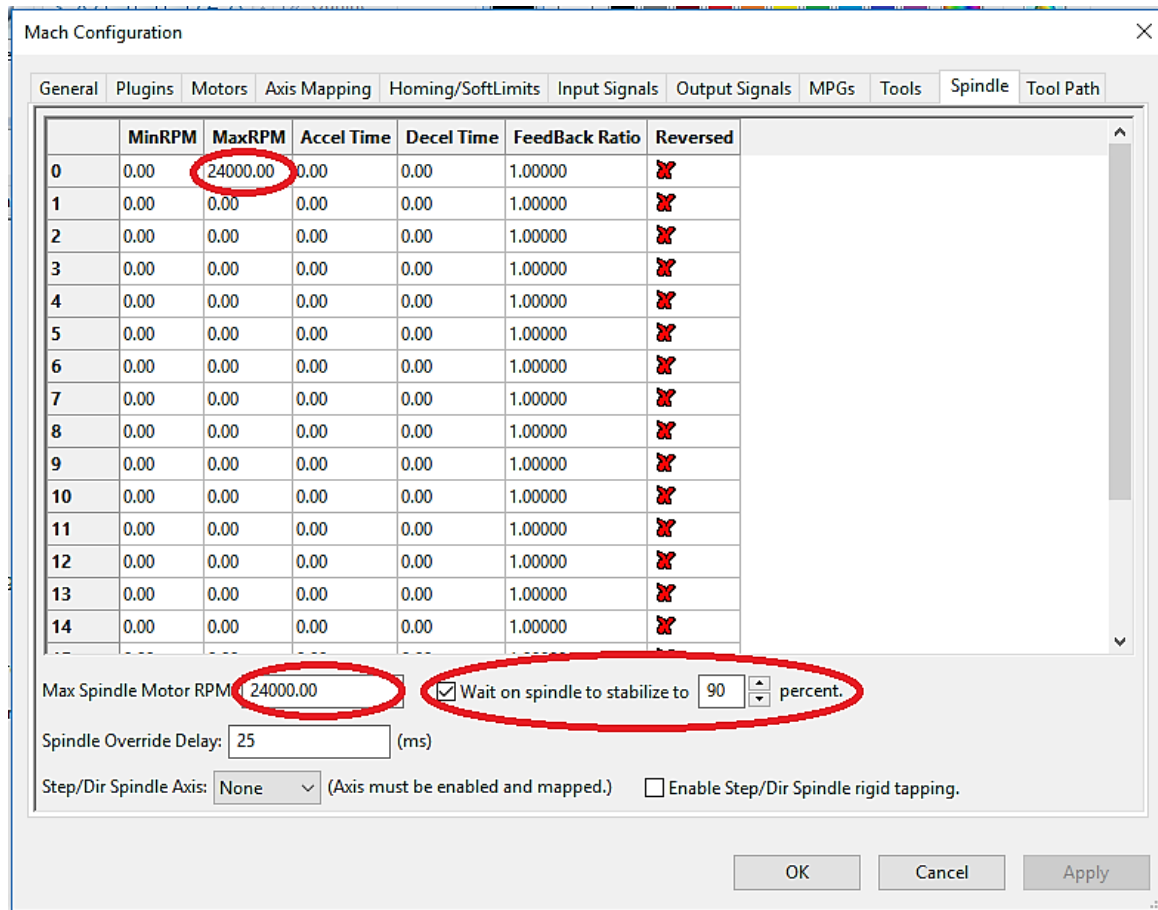
Spindle configuration screenshot

2. Go Configure / Plugins / ESS / Output Signals. In the Spindle Motor PWM, check Use Spindle Motor PWM and Spindle Motor Dir. Under Pulley Ratios set the pulley ratios of the machine.



Output Signals screenshot

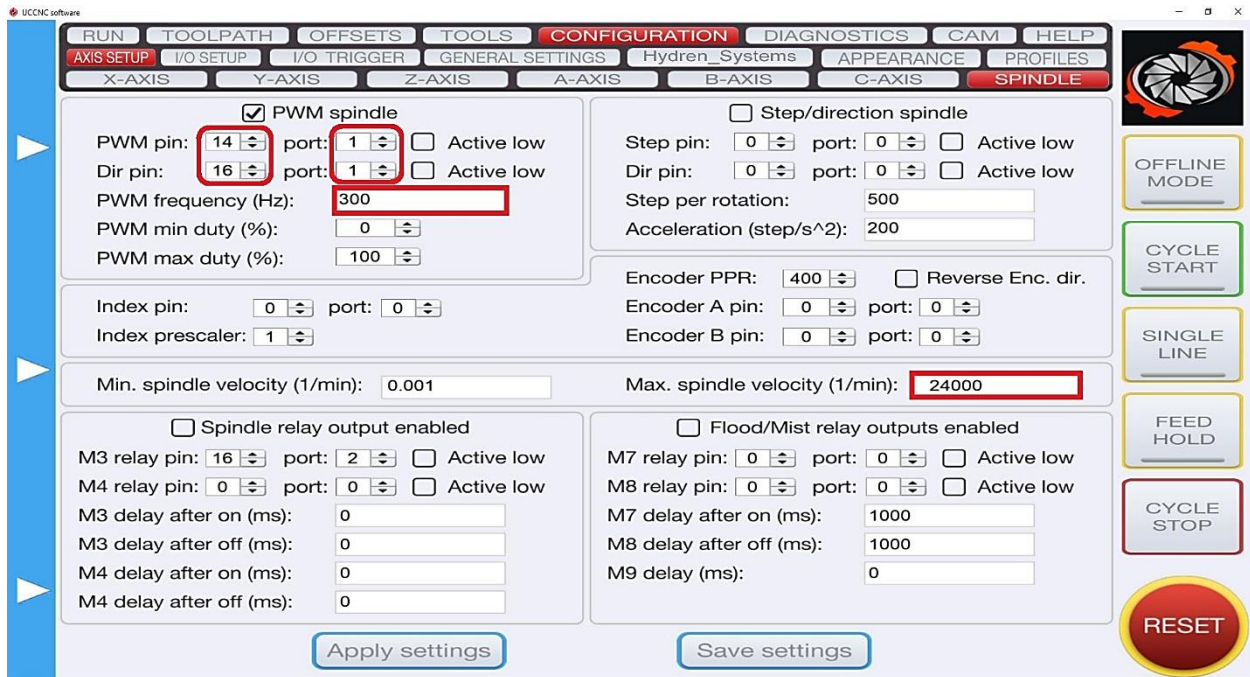
3. Go Configure / Mach / Spindle. Maximum for Velocity, Check use Wait on spindle to stabilize the spindle.



Spindle Velocity screenshot

9.2 For configuring UCCNC+UC300+C76 follow these steps:

Go to CONFIGURATION / AXIS SETUP / SPINDLE. Check use PWM spindle and select the Port, Pins, Velocity and Frequency.



The screenshot displays the UCCNC software interface, specifically the SPINDLE configuration window. The window is divided into several sections for configuring the spindle's operation. The top navigation bar includes tabs for RUN, TOOLPATH, OFFSETS, TOOLS, CONFIGURATION, DIAGNOSTICS, CAM, and HELP. The CONFIGURATION tab is active, and the SPINDLE sub-tab is selected. The SPINDLE configuration section includes the following settings:

- PWM spindle:** ☒ PWM spindle. PWM pin: 14, port: 1, Active low: ☐. Dir pin: 16, port: 1, Active low: ☐. PWM frequency (Hz): 300. PWM min duty (%): 0. PWM max duty (%): 100.
- Step/direction spindle:** ☐ Step/direction spindle. Step pin: 0, port: 0, Active low: ☐. Dir pin: 0, port: 0, Active low: ☐. Step per rotation: 500. Acceleration (step/s²): 200.
- Encoder settings:** Encoder PPR: 400, Reverse Enc. dir.: ☐. Encoder A pin: 0, port: 0. Encoder B pin: 0, port: 0.
- Index settings:** Index pin: 0, port: 0. Index prescaler: 1.
- Min. spindle velocity (1/min):** 0.001.
- Max. spindle velocity (1/min):** 24000.
- Spindle relay output enabled:** ☐ Spindle relay output enabled. M3 relay pin: 16, port: 2, Active low: ☐. M4 relay pin: 0, port: 0, Active low: ☐. M3 delay after on (ms): 0. M3 delay after off (ms): 0. M4 delay after on (ms): 0. M4 delay after off (ms): 0.
- Flood/Mist relay outputs enabled:** ☐ Flood/Mist relay outputs enabled. M7 relay pin: 0, port: 0, Active low: ☐. M8 relay pin: 0, port: 0, Active low: ☐. M7 delay after on (ms): 1000. M8 delay after off (ms): 1000. M9 delay (ms): 0.

Buttons at the bottom include "Apply settings", "Save settings", and a large red "RESET" button. On the right side, there are buttons for "OFFLINE MODE", "CYCLE START", "SINGLE LINE", "FEED HOLD", and "CYCLE STOP".

Spindle Setup screenshot

10.0 TIPS AND RECOMMENDATIONS:

Before applying power to the system, use a multimeter to verify isolation of the grounds. That is that the negative of the +5vdc that power the board are not common with the negative of the speed controller or VFD. You do that by setting the multimeter in the “beep mode”, then with each lead touch each ground that you want to verify.

After configuring the Mach4, these steps should be followed.

Step 1. Ensure that all external power sources are set to **OFF**.

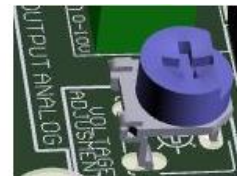
Step 2. Connect the power supply to the Power Inputs Terminals. If a RJ45 is used to wire the power lines from the C32S or C35S, don't use those terminals and connect the *RJ45 for inputs signals*.

Step 3. Connect the input signals to the Inputs Connectors. If a RJ45 is used to wire the signal from the C32S or C35S, don't use those terminals.

Step 4. Turn on the external supplies.

Step 5. Connect a multimeter in the analog outputs connectors and make a fine tune to this output:

Make sure that when you reach the max speed in the control software you get 10VDC out. This voltage can vary depending on many things, including the electrical properties of parallel port or breakout board you are using, the length of the step pulse your software is delivering, and the normal hi or low status of your step pin. Play with the fine tuning pot in the card, the normally hi or low status of your pin, and the pulse width.



↑
**pot for fine tuning
the analog
0-10VDC output**

Step 6. Turn off the external supplies.

Step 7. Connect the analog output and external Relay contacts.

Step 8. Turn on the external power supplies.

11.0 REPLACING A POTENTIOMETER:

For Replacing Potentiometer go to

http://cnc4pc.com/Tech_Docs/Replacing%20a%20Potentiometer.pdf

Top view of the CNC 41S Rev 1.1 PCB. The board is green with various components and labels. Dimensions are provided in inches. Key labels include: CNC 41S Rev 1.1, CNC 41S, TO BOARD, UFD ANALOG, REL1, REL2, SELECT/CONFIG UFD, COM1 REL1, COM1 REL2, MODE, FOR BOTH JUMPERS, 1-2=INT, 2-3=US, OPTIONAL UFD OUTPUT ANALOG, and various connector labels like 5V, GND, and others. The board has four mounting holes at the corners.

Dimensions (inches):

- Overall width: 3.94
- Overall height: 4.06
- Mounting hole diameter: 3.81
- Distance between mounting holes (horizontal): 100.076
- Distance between mounting holes (vertical): 52.959
- Distance from mounting hole to edge (horizontal): 3.81
- Distance from mounting hole to edge (vertical): 4.06

Disclaimer:

User's Manual